

Problem

In a balanced Δ -Y circuit, $V_{ab} = 440\angle 15^\circ$ and $Z_Y = (12 + j15) \Omega$. Calculate the line currents.

Step-by-step solution

Step 1 of 4

Convert the delta connected source to a wye connected source.

$$\begin{aligned} V_{an} &= \frac{V_{ab}}{\sqrt{3}} \angle -30^\circ \\ &= \left(\frac{440\angle 15^\circ}{\sqrt{3}} \right) \angle -30^\circ \\ &= 254\angle -15^\circ \text{ V} \end{aligned}$$

Therefore, the value of V_{an} is $254\angle -15^\circ \text{ V}$.

Step 2 of 4

Calculate the value of the line current I_a .

$$I_a = \frac{V_{an}}{Z_Y}$$

Substitute $254\angle -15^\circ \text{ V}$ for V_{an} and $12 + j15 \Omega$ for Z_Y .

$$\begin{aligned} I_a &= \frac{254\angle -15^\circ}{12 + j15} \\ &= \frac{254\angle -15^\circ}{19.21\angle 51.34^\circ} \\ &= 13.224\angle -66.34^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_a is, $13.224\angle -66.34^\circ \text{ A}$.

Compare $13.224\angle -66.34^\circ \text{ A}$ with $I_p \angle \theta^\circ$.

The values of I_p and θ° are 13.224 A and -66.34° respectively.

Step 3 of 4

Calculate the value of the line current I_b .

$$I_b = I_p \angle \theta^\circ - 120^\circ$$

Substitute 13.224 A for I_p and -66.34° for θ° .

$$\begin{aligned} I_b &= 13.224\angle -66.34^\circ - 120^\circ \\ &= 13.224\angle -186.34^\circ \\ &= 13.224\angle +173.66^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_b is, $13.224\angle +173.66^\circ \text{ A}$.

Step 4 of 4

Calculate the value of the line current I_c .

$$I_c = I_p \angle \theta^\circ + 120^\circ \text{ A}$$

Substitute 13.224 A for I_p and -66.34° for θ° .

$$\begin{aligned} I_c &= 13.224 \angle -66.34^\circ + 120^\circ \\ &= 13.224 \angle 53.66^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_c is, $13.224 \angle 53.66^\circ \text{ A}$.